

PIYUSH KUMAR PRAJAPATI

Ghazipur, Uttar Pradesh, India

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EDUCATION

Mahatma Jyotiba Phule Rohilkhand University

Bachelor of Technology (B.Tech) – Electronics & Communication Engineering

Bareilly, Uttar Pradesh

Expected Graduation: 2027

TECHNICAL SKILLS

HDL: Verilog HDL

EDA Tools: Xilinx Vivado, Cadence Virtuoso

Concepts: RTL Design, FPGA Design, Digital Logic, FSM, Timing Analysis, CMOS Design

EXPERIENCE

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

Summer Intern – Digital VLSI Design Flow using FPGA

Punjab, India

June 2026 – July 2026

- Designed and verified RTL modules using Verilog HDL in Xilinx Vivado.
- Performed simulation, synthesis, implementation, and timing analysis for FPGA-based designs.
- Implemented an FSM-based Traffic Light Controller and an 8-bit Dadda Multiplier with a Kogge-Stone Adder.
- Designed CMOS circuits including a 6T SRAM Cell, Two-Stage Op-Amp, and VCO using Cadence Virtuoso.

PROJECTS

High-Speed Dadda Multiplier with Custom Kogge-Stone Adder | Verilog HDL, Xilinx Vivado

- Implemented an 8-bit Dadda multiplier integrated with a custom Kogge-Stone adder.
- Implemented complete RTL architecture in Verilog HDL and verified functionality through simulation and comprehensive testbenches.
- Performed synthesis, implementation, timing analysis, and FPGA resource utilization using Xilinx Vivado.
- Compared timing performance with conventional multiplier architectures to evaluate optimization benefits.

FPGA-Based Traffic Light Controller | Verilog HDL, FSM, Basys-3 FPGA

- Designed a finite state machine (FSM) based four-way traffic light controller using Verilog HDL.
- Implemented clock division logic for real-time signal transitions on the Basys-3 FPGA development board.
- Verified functionality using simulation and deployed the design on FPGA using Xilinx Vivado.
- Applied XDC constraints and validated functionality on the Basys-3 FPGA.

CMOS Circuit Design using Cadence Virtuoso

- Designed transistor-level CMOS circuits including a 6T SRAM Cell, Two-Stage Operational Amplifier, and Voltage-Controlled Oscillator (VCO).
- Performed schematic design, functional verification, and analyzed circuit-level behaviour using Cadence Virtuoso.
- Simulated and verified CMOS circuits using Cadence Virtuoso.

CERTIFICATIONS

- Digital Design with Verilog – IIT Guwahati (NPTEL)
- Summer Internship in Digital VLSI Design Flow using FPGA – Dr. B. R. Ambedkar National Institute of Technology (NIT), Jalandhar